

HALTEN FYR, Norway

WMO: 012400

Lat: 64.173N

Lon: 9.406E

Elev: 16

StdP: 101.13

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-6.4	-4.5	-11.1	1.4	-5.3	-9.8	1.6	-3.9	21.0	1.1	19.9	2.2	9.1	110	0.903

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.9	18.7	15.5	17.4	14.6	16.4	13.9	15.9	18.1	15.1	17.0	14.2	16.0	5.5	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.8	10.5	17.1	14.0	10.0	16.4	13.2	9.5	15.5	44.7	18.3	42.3	17.0	40.1	16.1	18.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.1	17.9	16.1	DB	-7.1	21.5	3.1	1.5	-9.3	22.6	-11.2	23.5	-12.9	24.4	-15.2	25.5	(4)
(5)			WB	-8.4	16.9	2.9	1.4	-10.5	17.9	-12.2	18.7	-13.8	19.5	-16.0	20.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.4	2.2	2.2	2.9	5.3	8.0	10.5	13.2	13.8	12.2	8.5	5.7	3.6		
	DBStd	4.86	2.94	3.24	2.68	2.18	2.45	2.02	2.18	1.79	1.93	2.49	3.01	3.12		
	HDD10.0	1304	241	219	219	143	71	17	1	1	4	59	131	199		
	HDD18.3	4000	499	452	478	392	319	233	159	139	184	306	380	458		
	CDD10.0	347	0	0	0	1	10	33	101	119	69	12	1	0		
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH27.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	7.9	10.5	9.5	8.9	7.0	6.1	6.0	5.6	5.7	7.2	8.9	9.3	10.4		
(15) Precipitation	PrecAvg	897	96	87	81	52	45	59	53	65	89	92	89	96		
	PrecMax	1049	159	170	195	115	68	124	113	137	214	150	159	187		
	PrecMin	701	6	12	29	8	18	15	19	21	32	16	6	43		
	PrecStd	106	42	38	40	26	14	24	22	29	48	33	50	37		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.3	7.7	8.6	13.2	16.5	17.7	21.8	20.0	17.1	14.6	11.7	9.5		
		MCWB	6.7	6.4	6.1	9.2	12.1	13.5	17.2	16.2	13.6	11.7	9.8	7.8		
	2%	DB	7.6	7.2	7.5	10.8	14.2	15.7	19.3	18.4	16.1	13.4	11.0	8.8		
		MCWB	6.1	5.8	5.8	7.6	10.8	12.3	16.0	15.3	13.0	10.7	9.2	7.2		
	5%	DB	7.0	6.6	6.9	9.1	12.6	14.5	17.7	17.4	15.3	12.6	10.2	8.1		
		MCWB	5.6	5.2	5.5	6.7	10.0	12.0	15.0	14.7	12.7	10.2	8.3	6.4		
	10%	DB	6.3	6.0	6.4	7.9	11.3	13.4	16.4	16.5	14.7	11.7	9.3	7.4		
		MCWB	4.8	4.6	5.0	6.0	9.1	11.4	14.3	14.1	12.4	9.6	7.4	5.7		
	0.4%	WB	7.5	6.9	7.1	9.3	12.5	14.3	17.9	16.5	14.2	12.2	10.5	8.6		
		MCDB	8.0	7.5	7.5	12.1	15.5	16.8	20.9	19.1	15.6	13.6	11.4	9.1		
	2%	WB	6.6	6.1	6.5	8.1	11.2	12.9	16.4	15.7	13.7	11.1	9.5	7.6		
		MCDB	7.4	6.7	7.2	10.4	13.6	14.7	18.9	17.8	15.1	12.9	10.7	8.5		
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	5.7	5.4	5.7	7.1	10.3	12.1	15.4	15.1	13.3	10.5	8.6	6.6		
		MCDB	6.8	6.3	6.7	8.5	12.3	13.9	17.4	17.0	14.8	12.2	10.0	7.8		
	10%	WB	4.9	4.8	5.1	6.4	9.3	11.5	14.5	14.3	12.8	9.8	7.6	5.8		
		MCDB	6.2	5.9	6.3	7.6	11.0	13.1	16.4	16.1	14.4	11.6	9.3	7.2		
	5% DB	MDBR	2.8	2.7	2.8	3.0	3.3	3.0	3.1	2.9	2.6	2.4	2.6	2.8		
		MCDBR	3.0	2.6	2.9	4.1	5.3	4.8	5.3	4.2	3.2	2.8	2.8	3.4		
(36) Mean Daily Temperature Range	5% WB	MCWBR	3.1	2.9	2.8	2.6	3.1	3.0	3.1	2.5	2.3	2.5	2.8	3.7		
		MCDBR	2.9	2.6	2.8	3.5	5.1	4.3	4.9	3.9	3.0	2.7	2.8	3.4		
	5% WB	MCWBR	3.2	3.0	2.9	2.5	3.2	3.0	3.1	2.6	2.4	2.6	3.1	3.7		
		MCDBR	3.2	3.0	2.9	2.5	3.2	3.0	3.1	2.6	2.4	2.6	3.1	3.7		
(40) Clear-Sky Solar Irradiance	taub		0.382	0.317	0.340	0.359	0.353	0.356	0.363	0.352	0.341	0.315	0.353	0.360		
	taud		2.564	2.242	2.326	2.375	2.458	2.485	2.491	2.517	2.515	2.388	2.517	2.446		
	Ebn at Noon		296	586	743	816	859	863	845	822	752	595	294	120		
	Edh at Noon		27	58	80	95	95	95	92	82	67	49	24	15		
(44) All-Sky Solar Radiation	RadAvg		0.11	0.56	1.74	3.46	4.84	5.14	4.73	3.64	2.13	0.89	0.20	0.05		
	RadStd		0.02	0.09	0.24	0.30	0.23	0.52	0.45	0.34	0.26	0.13	0.04	0.00		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page